

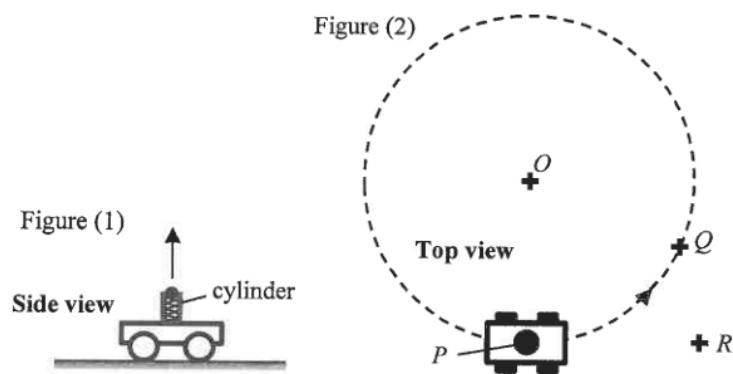
- *11. A car of weight W travels with constant speed on a circular humpback bridge. At the moment when the car passes the top of the bridge, the normal reaction acting on it by the bridge is R . What is the centripetal force acting on the car at that moment ?



Side view

- A. $W + R$
- B. $W - R$
- C. W
- D. R

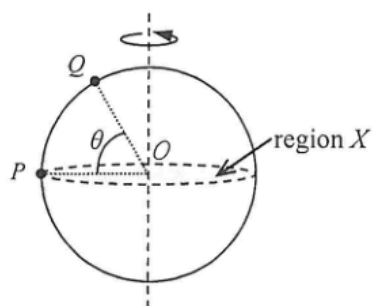
- *9. A trolley with a spring-loaded launcher in Figure (1) can project a ball vertically upward. The trolley travels on the ground at a constant speed in a horizontal circle (with centre O) as shown in Figure (2).



The ball is projected when the trolley is at point P . After some time, the ball falls back to the ground as the trolley just reaches point Q . Neglecting air resistance, where would the ball land ?

- A. O
- B. P
- C. Q
- D. R

- *9. Particles P and Q are fixed on the surface of a sphere rotating about a vertical axis passing through the centre O of the sphere as shown. The horizontal shaded region X divides the sphere into two halves. P is at the edge of region X while Q is at an angle of elevation θ above region X .

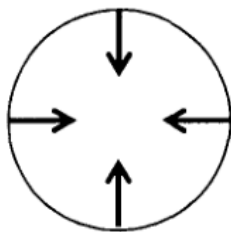


Find the ratio of the centripetal acceleration of P to that of Q .

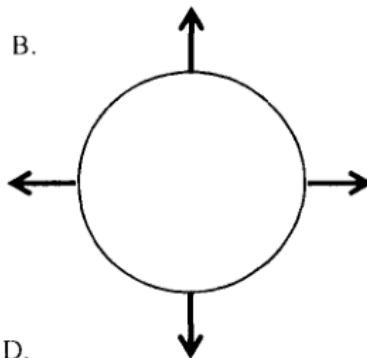
- A. $1 : \cos \theta$
- B. $1 : \sin \theta$
- C. $\cos \theta : 1$
- D. $\sin \theta : 1$

- *9. A particle is moving clockwise (top view) in a horizontal circle with uniform speed. Which top view diagram below correctly shows the net force acting on the particle at various positions ?

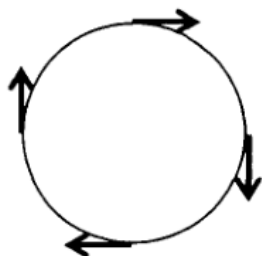
A.



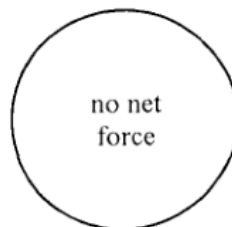
B.



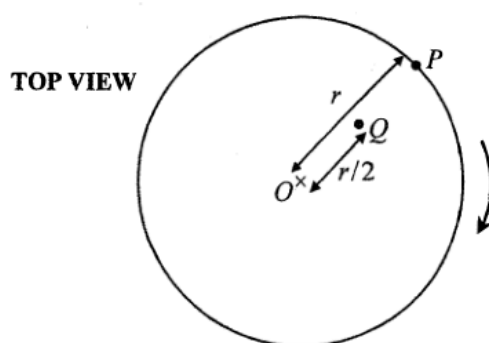
C.



D.



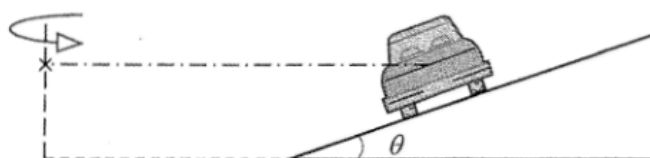
- *13. Particles P and Q are fixed at distances r and $r/2$ respectively from the centre O of a horizontal circular platform which is rotating uniformly as shown.



The ratio of the acceleration of P to that of Q is

- A. 1 : 2.
- B. 2 : 1.
- C. 1 : 4.
- D. 4 : 1.

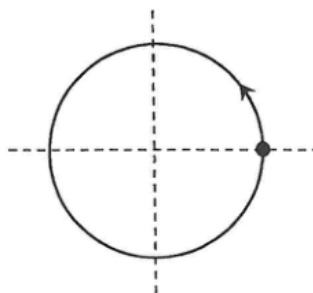
*10.



The figure shows the rear view of a car of mass m which travels along a circular road banked with an angle θ to the horizontal. The car moves at a certain speed such that it experiences **no frictional force along the inclined surface**. Which of the following represents the centripetal force on the car ?

- A. $mg \sin \theta$
- B. $mg \sin \theta \cos \theta$
- C. $\frac{mg \cos \theta}{\sin \theta}$
- D. $\frac{mg \sin \theta}{\cos \theta}$

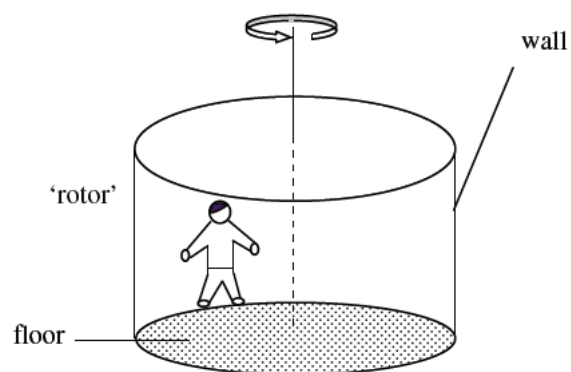
*12.



The figure shows a particle performing uniform circular motion. Which of the following correctly indicates the directions of its velocity and acceleration at the instant shown ?

- | | velocity | acceleration |
|----|-----------------|---------------------|
| A. | ↑ | ← |

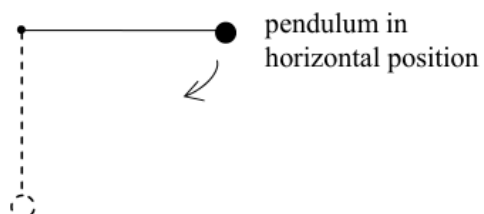
*15.



A man is rotating with constant speed inside a cylindrical 'rotor' and he remains pressed against the wall. The floor of the 'rotor' is smooth. Which of the following forces provides the centripetal force for the man ?

- A. the weight of the man
- B. the frictional force from the wall
- C. the normal reaction from the wall
- D. the supporting force from the floor

*14.



A simple pendulum is held at rest in a horizontal position. It is then released with the string taut. Which statement about the tension in the string is **not correct** when the pendulum reaches its vertical position ?

- A. The tension equals the weight of the pendulum bob in magnitude.
- B. The tension attains its greatest value.
- C. The tension does not depend on the length of the pendulum.
- D. The tension depends on the mass of the pendulum bob.